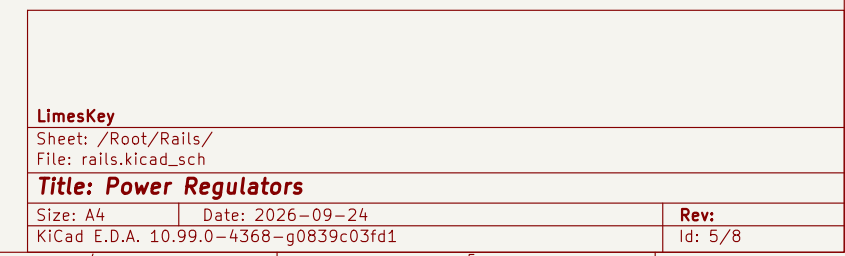


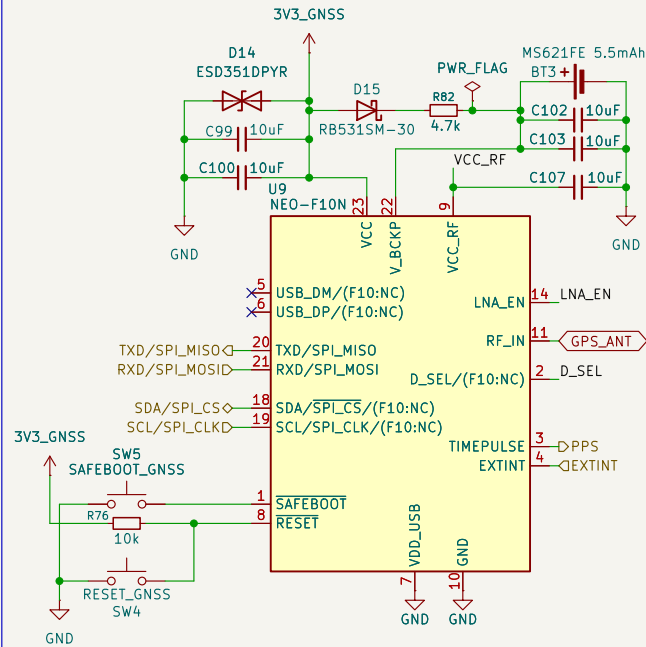
3V3 Buck

The schematic diagram illustrates a 3V3 Buck converter circuit. The central component is the LM61460AASQRJRRQ1 IC (U13). The input side (VIN1, VIN2) is connected to VSYS and filtered by capacitors C122, C123, and C124. The feedback network (FB) includes a divider (R104, R105, R109) and a junction point (JP16) connected to VSYS. The output side (SW, CBOOT, RBOOT) is connected to VSYS and filtered by capacitors C131, L4, C135, C141, C143, C190, C191, and C192. The PWR_FLAG output is connected to the output of the converter. The circuit is powered by a 3V3 supply (PGOOD_3V3) and includes a 3V3 output (3V3).



Id: 5/8

High to enable I2C/UART, low to enable SPI
Only for NEO-M(♯)N



Sparkfun Schematic: https://cdn.sparkfun.com/assets/1/0/a/8/a/SparkFun_GPS_NEO_M9N_SMA_UPDATE.pdf
 U-Blox Datasheet: https://content.u-blox.com/sites/default/files/NEO-M9N_Integrationmanual_UBX-19014286.pdf

The schematic diagram illustrates the power management section of the TQ2401 module, featuring two LDOs (U10 and U11) and various passive components.

U10: TPS7A2033PDBVR (High PSRR & Low Noise LDO)

- Input:** Connected to the 3V3 pin via a 0Ω resistor (R134) and a 2.2μF capacitor (C104).
- EN (Pin 1):** Connected to the 3V3 pin.
- OUT (Pin 5):** Connected to the 3V3_GNSS pin via a 10μF capacitor (C130).
- NC (Pin 4):** Connected to GND.
- GND (Pin 2):** Connected to GND.

U11: TPS259474LRPWR (Optional module power input protection)

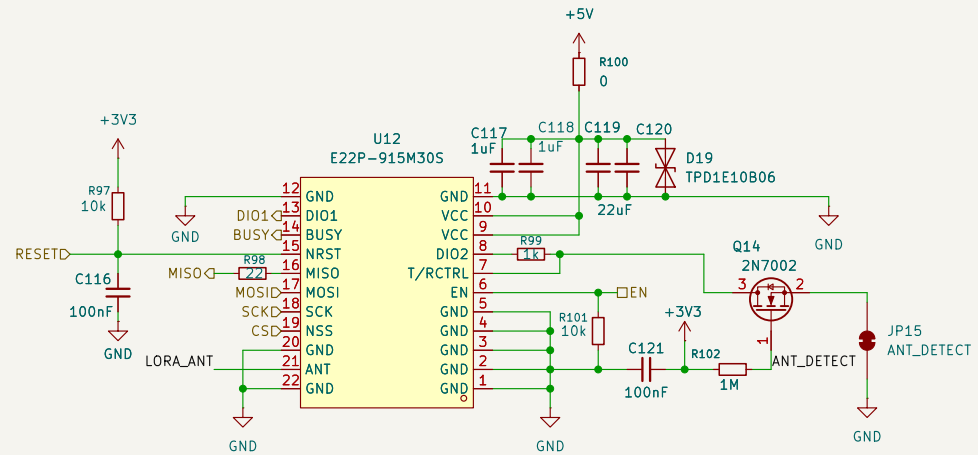
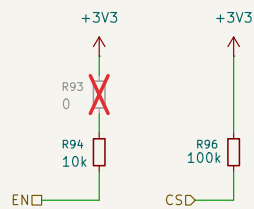
- IN (Pin 5):** Connected to the 3V3_GNSS pin.
- EN/UVLO (Pin 1):** Connected to GND.
- OVLO/OVCSEL (Pin 2):** Connected to GND.
- PGTH/FLT (Pin 4):** Connected to the 3V3_GNSS pin.
- PG/AUXOFF (Pin 3):** Connected to GND.
- ITIMER (Pin 6):** Connected to GND.
- DVDT (Pin 7):** Connected to GND.
- GND (Pin 8):** Connected to GND.
- ILIM (Pin 9):** Connected to GND.

Passive Components:

- Resistors:** R84 (150k, 0.5%), R85 (215k, 0.1%), R87 (113k, 0.1%), R88 (6.65k, 1%), R89 (200k, 1%), R90 (301k), R91 (100k).
- Capacitors:** C104 (2.2μF), C130 (10μF), C106 (10μF), C108 (2.2nF), C109 (2.2nF), C110 (10μF), C111 (10μF).
- Diode:** D17 (TPD1E10B06).
- Jumper:** JP13 (DISABLE_GNSS_PROT).

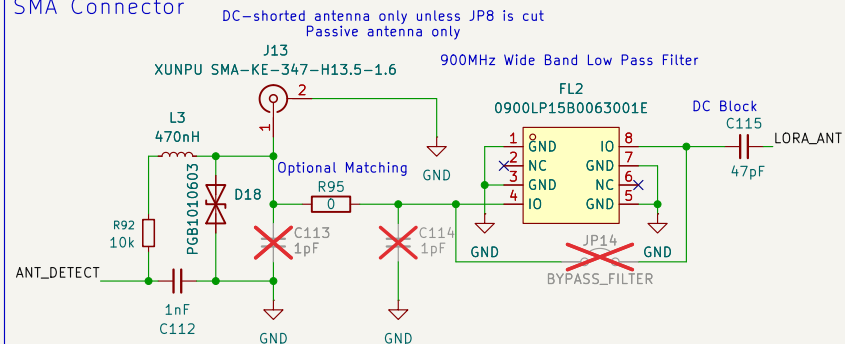
Output: The 3V3_GNSS pin is connected to the 3V3 pin via a jumper (JP13).

Rev: 1.1
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Ensure you set DIO2 as RF Switch in program
https://jgromes.github.io/RadioLib/class_s_x126x.html#ae46e08d579f4acbad029b4cd4f4ffaf

SMA Connector



LimesKey

Sheet: /Root/LoRa/
 File: lora.kicad_sch

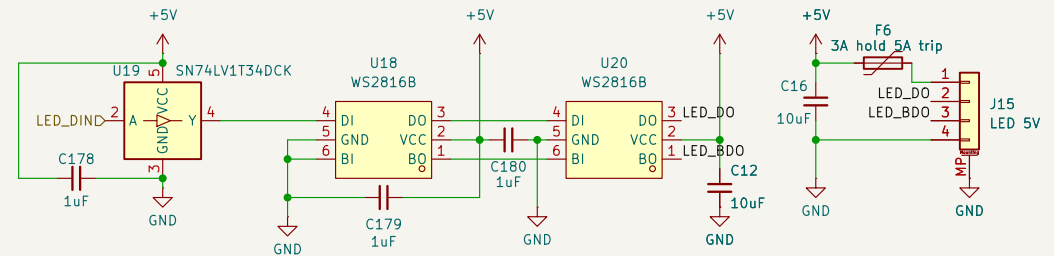
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Size: A4 Date: 2026-08-24
 KiCad E.D.A. 10.99.0-4368-g0839c03fd1

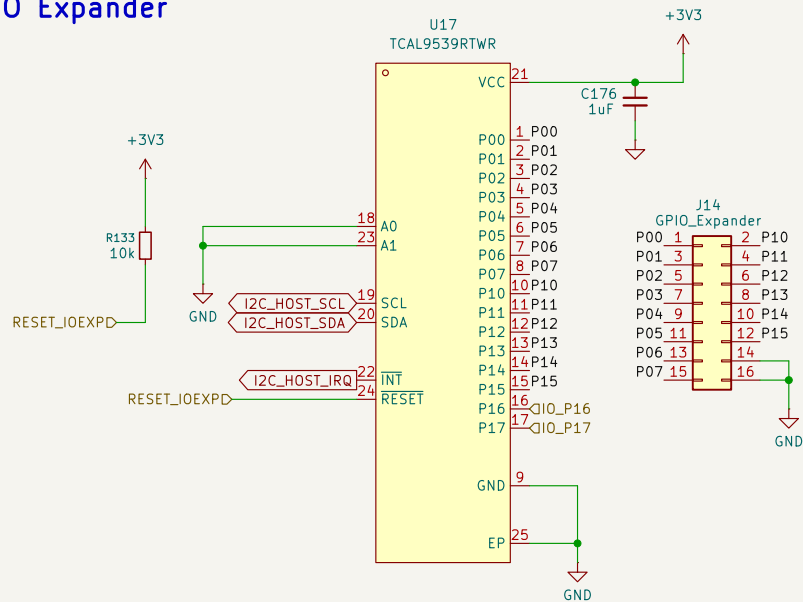
Rev:
 Id: 7/8

Individually Addressable RGB LEDs

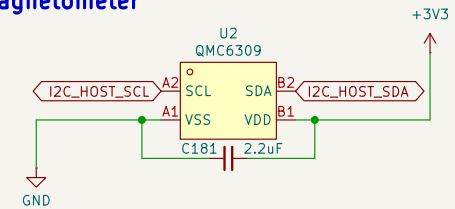
For use with WS2813/WS2816



GPIO Expander



Magnetometer



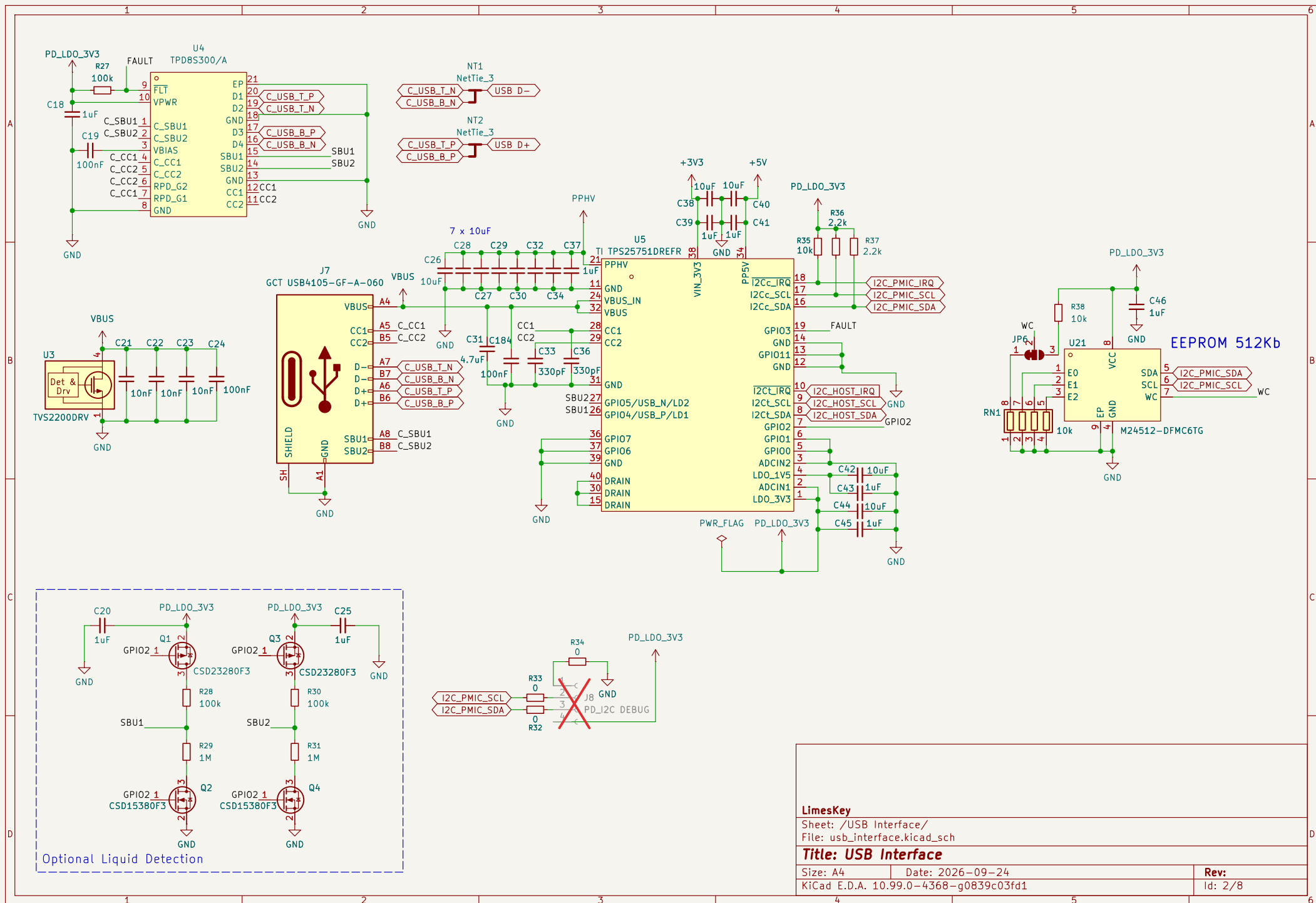
LimesKey

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File: peripherals.kicad_sch

Title: GPIO Expander / LEDs / Magnetometer

Size: A4 Date: 2026-09-24
KiCad E.D.A. 10.99.0-4368-g0839c03fd1

Rev:
Id: 8/8



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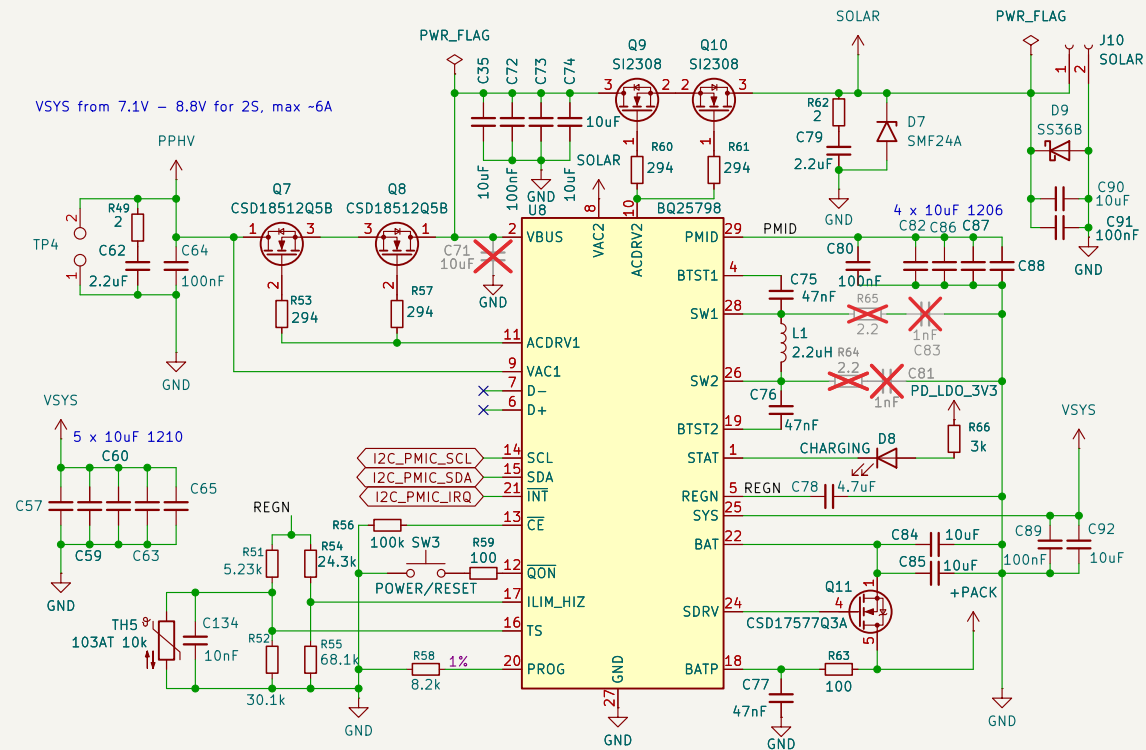
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Title: USB Interface

Size: A4 Date: 2026-09-24
KiCad E.D.A. 10.99.0-4368-g0839c03fd1

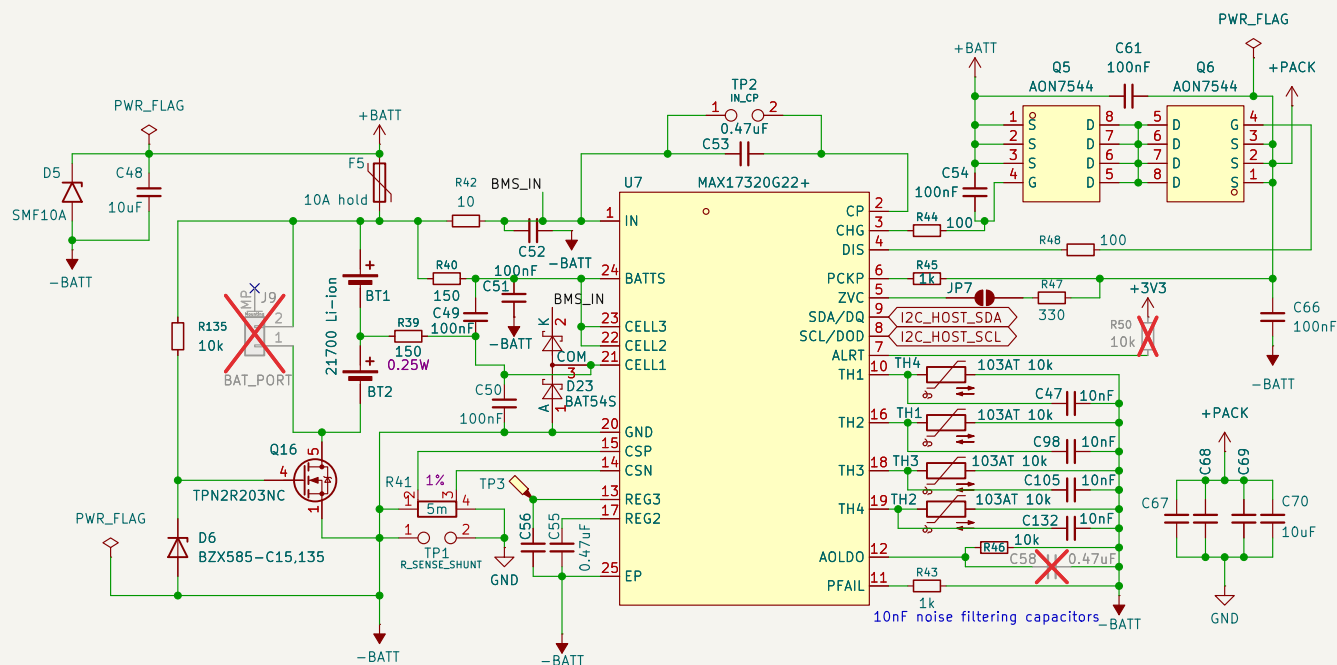
Rev:
Id: 2/8

Capacitors voltage ratings specified,
generally >25V – use XR7/C0G



Id: 3/8

Battery Management System (BMS)



LimesKey

Sheet: /BMS/

File: bms.kicad_sch

Title: Battery Managment System (BMS)

Size: A4 Date: 2026-09-22

KiCad E.D.A. 10.99.0-4368-g0839c03fd1

Rev:

Id: 4/8